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Coal mining: Risks to fish population and human communities

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Abstract- Coal mining is a major contributor to environmental pollution, releasing heavy metals, nitrate, sulphate, dissolved, and suspended solids as well as other harmful pollutants into aquatic ecosystems. Significant discharge of untreated coal mining effluent into the aquatic environment contributes to a variety of toxic effects on aquatic flora and fauna. These pollutants exert profound effects on freshwater fish, altering biochemical and antioxidant enzyme parameters, causing histopathological alterations, thereby impairing growth and reproduction. Bioaccumulation of heavy metals in fish further poses significant human health risks, including neurological disorders, kidney and liver dysfunction, and carcinogenicity. This review provides current research from major Indian coalfields and presents comprehensive summaries of alterations in fish health parameters, including physiological mechanisms. Moreover, this emphasizes the need for environmental monitoring, regulatory enforcement, and mitigation strategies to protect aquatic ecosystems and human populations near coal mining regions.

Keywords: Coal Mining, Fish, Human Health, Biochemical, Antioxidant Enzymes

INTRODUCTION

In fast moving, new era of science and technology, the world has observed tremendous transformations in all aspects of life. At the heart of this tremendous transformation rapid industrialization is taking place across the globe. The coal mining in India is a major industrial activity which contributes significantly to the economy of country.[1] Amongst the different fossil fuels, the coal is the most abundantly available and cheapest resource of energy.[2-3] Coal alone fulfils 29.6% of global primary energy requirement. About 42% of the world's electricity is generated from coal. The global coal consumption has increased by 46% during 2001 to 2010.[4] India is the third largest coal producer in the world after China and USA.[5] The state of Jharkhand, located in North-eastern India tops the list with the maximum coal reserves of 80 billion tonnes.[6]

Among the two mining methods of coal extraction underground mining is less detrimental than surface or opencast mining.[7-8] During the process of open cast mining, a variety of rock types with different compositions are exposed to atmospheric condition and undergo accelerated weathering.[9-10] During the mining operations, huge quantities of underground mine water are pumped out from underground sumps and surface mining pits.[11] This effluent from coal mines contains high load of total suspended solids (TSS), total dissolved solids (TDS), nutrients and heavy metals.[12-17] The surrounding water resources get affected when this effluent is discharged into the aquatic eco-system of the area due to lack of proper water management plan in most mines.[18-22] Thus,

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deterioration of water quality as well as decline in biodiversity in the water bodies around mining areas is apparent.[23-26]

Upgupta and Singh, (2017)[27] studied the adverse effects of coal mining in Jaintia Hills in Meghalaya on the water quality and revealed that the water is completely unfit for agriculture and human consumption, and even highly toxic to the native flora and fauna. Talukdar *et al.*, (2017)[28] reported that coal mining activities in East Garo hills, Meghalaya is posing serious threats to the fish and fisheries of the Simsang River.

In aquatic ecosystems, fishes may be considered as bioindicators in determining the impact of water contaminants.[29] Several characteristics make them excellent experimental models for toxicological research such as they respond with a greater sensitivity to changes in the aquatic environment. Fish are especially vulnerable to these pollutants due to bioaccumulation, which affects physiological, biochemical, and genetic health and poses direct risks to human consumers.[30-31] Talukdar *et al.,* (2017)[28] have reported that pollution of rivers by coal mining activities induced stress on the freshwater fish *Channa punctatus*, making it weak and more vulnerable to diseases. Studies across Indian coalfields have documented multi-level impacts on fish, from enzyme alterations to genotoxicity, highlighting the need for integrated monitoring of aquatic health and public safety.

Even though coal mining and mixing of CME with aquatic systems are inevitable on account of their contribution to the economy of the country, it is necessary to look for ways to protect aquatic life and to prevent and control coal-mine water hazards. This is because the pollutants from coal mines have tremendous impact on the fish fauna and may completely wipe out the population that thrives therein in the long run.

MATERIALS & METHODS

The literature for this review was searched from PubMed and Google scholar, using the following keywords coal mining, human health, biochemical, antioxidant enzymes and histopathology in combination with fish. All included articles were cited along with their specific references.

DISCUSSION

i) Impact of Coal Mining on Biochemical Parameters in Fish

The effects of toxicants on fish are multidirectional and are manifested by numerous changes in the physiological and biochemical process of their organ system. The different biochemical parameters are sensitive for detecting potential adverse effects of metal accumulation. Coal mining effluents containing heavy metals and suspended solids are absorbed by fish through gills, skin, and ingestion, leading to accumulation in liver and muscle tissues. The biochemical profiles including the activities of various enzymes are used in fish toxicity tests and for field monitoring of aquatic ecosystem.[32-34] The coal mine effluent induced physiological and some biochemical abnormalities due to excessive accumulation of metals in fish tissues are reported by Bharti and Banerjee, (2014a,b).[24,25] According to Lakra *et al.,* (2022)[31] the changes observed in protein, glucose and lipid contents of CME-exposed fish appear to affect their nutritive value in addition to changes in the normal functioning.

ii) Impact of Coal Mining on Antioxidant Enzymes in Fish

A variety of xenobiotics including heavy metals exert their toxic effects on survival of fish by producing reactive oxygen species (ROS) that causes oxidative stress, cellular damage and lesions in tissues.[35,36] Coal mining pollutants generate ROS in fish tissues, overwhelming antioxidant defenses. SOD, CAT, and GPx neutralize superoxide and hydrogen peroxide; their inhibition leads to lipid peroxidation, DNA oxidation, and protein damage.[37,38] Mitochondrial dysfunction reduces ATP, impairing muscle contraction and metabolic processes. Indian species like *Clarias batrachus* demonstrate marked oxidative stress responses upon exposure, showing physiological impairment of gill, liver, and kidney function.[39]

iii) Impact of Coal Mining on Tissue Damage Marker Enzyme Activity in Fish

The presence of toxicants in aquatic media is reported to exert their effects at the cellular and molecular levels, which are exhibited in biochemical parameters. Metals like Pb, Cd, and Cr interfere with key enzymatic pathways such as transamination (AST, ALT) and phosphatase activity (ALP), affecting nitrogen metabolism and protein turnover. Hepatic cells metabolize xenobiotics but are vulnerable to metal-induced oxidative stress. Lipid metabolism is also disrupted due to peroxidative damage to membranes, resulting in altered cholesterol and triglyceride levels. Coal mining pollutants cause cell

membrane disruption in liver and kidney tissues, resulting in leakage of AST, ALT, and ALP.[30,31] Metals bind to sulfhydryl groups of enzymes, altering their activity and signaling apoptotic pathways. In Indian species, this enzymatic alteration correlates with histopathological lesions, providing early evidence of organ stress and impaired physiological homeostasis.[31]

iv) Impact of Coal Mining on Hemato-Biochemical Parameters in Fish

Fish blood is sensitive to pollution- induced stress hence the study of haematological and serological biochemistry of blood serves as a valuable tool to monitor stress caused by pollutants.[40,41] Coal mining pollutants affect hematopoiesis by disrupting erythrocyte and leukocyte production. Hemoglobin synthesis is impaired by metals leading to hypoxia. Leukopenia indicates immuno suppression, while elevated glucose, cortisol, urea, and creatinine reflect stress and renal dysfunction. Talukdar *et al.,* (2015)[42] have reported significant decrease in the content of haemoglobin (Hb), total erythrocyte count (TEC), packed cell volume (PCV) and mean corpuscular volume (MCV) values. They have also observed marginal variation in the level of mean corpuscular haemoglobin (MCH) and mean corpuscular haemoglobin concentration (MCHC) and increase in white blood corpuscles (WBC) count in fish *C. punctatus* inhabiting in coal mining affected areas of the Simsang River. Bharti and Banerjee, (2014b)[24] have also reported decreased values of red blood corpuscles (RBC), Hb, haematocrit (Hct), MCH and MCHC while increased in WBC and MCV after raw CME exposure. Alterations of blood cell numbers as well as biochemical changes were used as an index to assess health of fish during CME stress. Physiologically, these changes reduce oxygen transport, impair immune response, and compromise metabolic regulation.[30-31]

v) Impact of Coal Mining on Fish Histopathology

The histopathological techniques in environmental monitoring helps to visualize the alterations taking place at structural as well as chemical levels in the cells and tissues qualitatively.[43,44] Mylliemngap and Ramanujam, (2012)[45] reported morphological changes like fusion, distortion and loss of alignment of primary and secondary lamellae of the gills of *H. fossilis* exposed to coal mine effluent contaminated water of Rymbai River in Jaintia hills, Meghalaya, India. Histopathological changes occur in gills (lamellar fusion, epithelial necrosis), liver (hepatocyte necrosis, vacuolation), and kidney (tubular degeneration) due to oxidative stress and metal accumulation.[30] Talukdar *et al.,* (2017)[28] have also reported hepatic and renal histological alterations in fish *Tor tor* collected from Simsang River, contaminated with coal mining waste. Lakra *et al.,* (2019)[39] have revealed that fish inhabiting the CME-fed pond water near the Rajrappa coal mining complex were contaminated with substantially high levels of heavy metals, and consequently, their biochemical configurations and histomorphology of various vital organs were greatly altered.

vi) Impact of Coal Mining on Human Health

Each phase of coal's lifecycle (mining, disposal of contaminated water and tailings, transportation, washing, combustion, and disposing of post combustion wastes) produces pollutants that affect human health.[46-48] Underground coal mining is recognized to raise the risk of lung cancer, chronic obstructive pulmonary disease, and pneumoconiosis in terms of occupational exposures. These studies clearly show that residents in coal mining areas face a variety of health issues. These include various types of cancer, respiratory and cardiovascular conditions, kidney disease, developmental issues, depression, lower health-related quality of life, and a wide range of illness symptoms.[49-51]

CONCLUSION

The findings collectively indicate that using coal for large-scale energy production or domestic usage has significant negative effects on the environment and public health. The data unequivocally show how urgent it is to switch to greener and more sustainable energy sources to protect the health of our planet and the welfare of our communities. Based on increasing environmental pollution from coal mining and the rising need of raw coal in energy generation, a key research area for environmentalists is to design effective and eco-friendly waste water treatment strategies with minimum economic costs. This review will serve as a valuable resource for policymakers, industry stakeholders, researchers, and activists for making informed decisions and promoting cooperative efforts to address the environmental and health risks associated with coal mining and use.

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